

Impact Statement

SLM for Agriculture and Climate — agriculture-climate-slm-challenge

This project fine-tunes a small language model to generate concise, factsheet-grounded answers to smallholder farmers' agronomic questions, aiming to address a real gap: extension knowledge on crop disease, pests, soil health, water, and climate adaptation exists, but is not accessible in a conversational, on-demand form for farmers in low-connectivity, low-hardware environments across Africa's semi-arid, sub-humid, and highland zones.

The core value guiding this work is accuracy and trust over generic AI advice. General-purpose LLMs risk hallucinating application rates, timings, or practices that carry real economic and food-security consequences if acted on incorrectly for instance, an incorrect compost quantity or a wrong planting window can waste a season's inputs or labor. By grounding every answer in a specific, matched extension document rather than open-ended generation, and by evaluating outputs against short, specific reference answers, this design favors precision and traceability over persuasive-sounding but unverified text.

The potential benefits are meaningful if the approach matures beyond this benchmark: faster access to locally relevant advice for extension workers serving many farmers with limited time, a model small enough to eventually run offline on modest hardware, and a development pattern — metadata-grounded generation over a small, curated corpus — that other under-resourced agricultural contexts could adapt with their own vetted content.

However, real risks and limitations must be named rather than assumed away. The current dataset is entirely synthetic, generated without direct input from farmers, extension officers, or agronomic experts, and covers only 7 crop categories and 3 agro-zones. Deploying a model trained on this data as if it were verified advice would risk real harm: farmers acting on unvalidated recommendations, or communities outside the covered crops and zones being served poorly or not at all, widening rather than closing the information gap the project intends to address. There is also a dignity and autonomy dimension: any real deployment should position the model as a supplement to, not a replacement for, human extension expertise and farmers' own situated knowledge, so that trust in the tool does not come at the expense of farmers' agency to question or override its answers.

Over time, responsible development would mean validating factsheet content and phrasing with real extension workers and farmer groups before any deployment beyond a benchmark setting, expanding coverage to underrepresented crops and zones, and being transparent with any future users about the synthetic origin and current limitations of the underlying data. Accountability to the communities this tool is meant to serve requires treating today's results as a

promising prototype, not a finished, trustworthy advisory system.